

GROW computing scheme of work

GROW pathway • Core course, continuation and optional routes

The planned starting course is an eight-session Scratch maze, taught for 40 minutes a week. The main unit is AQA 71638 Programming with Scratch (unit 6), Level One.

[AQA 71638: Programming with Scratch \(unit 6\)](#)

Intent and intended learners

Develop confidence controlling a computer, expressing an idea through code, diagnosing a problem and improving a usable product. Assume a secondary GROW group with varied reading, attention, emotional regulation and prior computing experience. Adapt to the school's own pathway profile and individual access needs.

The curriculum route

Route	Time at one lesson a week	Product / role
Core maze course	8 sessions, 320 minutes	Playable maze with required AQA features
Scoring continuation	4 further sessions, 160 minutes	Two score variables and event-linked updates
Variables conversation	4 after basics; 5-6 for novices	Fictional interactive mission guide
Three-page website	8 sessions, 320 minutes	Local interest site or mini exhibition
Sound and visual feedback	2-3 after a working game	Refined feedback in the game
micro:bit comparison	6-8 with physical devices	Three counters and practical evaluation
Python bridge	Future 6-8, subject to readiness	Small text program; award to be verified

The optional routes are choices for later teaching, not additional projects to squeeze into the core eight weeks. The accompanying Eight-week Plan provides the detailed core lesson sequence. Weeks 1 and 2 have full teaching packs, each with Step by step, Main task and Challenge resources. The Eight-week Plan gives concrete adaptations for every core lesson.

Entry and completion

No prior Scratch experience is assumed. Confirm pointer/keyboard access, file handling and a workable prompt level in Weeks 1-2. Course completion means a working pupil-created game plus evidence of every unit outcome. A partially completed game receives a recorded next step rather than automatic unit completion.

UAS recognises completed learning outcomes. A unit labelled Level One is not a regulated Level 1 qualification or a GCSE equivalent. Confirm the centre's registration, evidence and submission arrangements with the school UAS lead.

Core course and AQA alignment

Unit 71638 • Level One • All six outcomes required

The descriptions below are planning paraphrases. Use the live AQA wording for final checking and the centre's official summary sheet.

No.	Required pupil achievement	Planned week	Evidence opportunity
1	Create maze, controllable object and finish sprites.	3	Pupil construction and identification
2	Create keyboard arrow-control scripts.	4	Pupil scripts and live directional test
3	Create a consequence when the player touches the maze.	5	Pupil script and observed contact response
4	Create a response when the game is won.	6	Pupil script and demonstrated finish
5	Add a feature that increases complexity.	7	Pupil explanation and changed gameplay
6	Test the game.	7-8	Observed tests and retest

The published evidence requirement is a summary sheet for every outcome. Saved projects, short test records and screenshots support the teacher's record but do not replace the official evidence process.

Progression across the eight sessions

Week	Curriculum knowledge and application
1	Events, block order and numeric values. Modify one movement value and save/reopen.
2	Key input, horizontal/vertical position and debugging. Construct initial controls.
3	Sprites and layout. Create separate maze, player and finish sprites.
4	Events and coordinate changes. Create reliable arrow controls in the maze.
5	Repeated checking and conditions. Create a consequence for touching the maze.
6	Win condition and starting state. Create a winning response and a playable restart.
7	Gameplay complexity. Add a feature that changes player decisions or actions.
8	Expected versus actual results. Test, improve, retest and review all outcomes.

Assessment boundaries

A maze drawn only as a backdrop does not meet the three-sprite outcome. Decoration alone does not meet the complexity outcome. Using a supplied working project does not show that a pupil created the required scripts. Give a later construction opportunity for any inherited assessed code.

Teaching assessment and access

The GROW chassis guides the whole class and every route

Choose support from the current task

Use the same lesson goal and eight named stages. At We Do 2 · Task, check whether each pupil can locate the next block or control, choose the next action and predict its effect. Match the task route to that evidence and the pupil's access needs. Recheck during the task; pupils may move between routes and may need different support for coding and writing.

Route	What changes	What remains required
Step by step	One action at a time; larger print; visual block and number references; spoken or pointed responses.	The shared coding goal, practical demonstration and every relevant AQA outcome.
Main task	A short task sequence; reference available; pupil chooses and tests their own values or code.	Individual construction where required, explanation, testing and saving.
Challenge	A new constraint, awkward test or reasoned comparison after the shared goal.	A reliable core program and evidence of the compulsory features.

Adult prompts and access

Wait; invite self-check; point to the current visual step; then give a general verbal prompt. Model again when needed and record direct help. Fade one prompt after success. Provide a quiet help signal, a short regulation break where needed and a visible return step. Keep language short, around reading ages 7-11, and avoid lengthy copying.

Accept speaking, pointing, drawing or an adult scribing the pupil's exact words. For motor access an adult may operate at the pupil's direction; record precisely who decided and who acted, and confirm the evidence method with the UAS lead. Do not infer code creation from adult operation or inherited code. Pair work needs a keyboard-role swap and an individual demonstration.

Pupil voice with a visible response

End every lesson with "What I said, and what it changed". Give time and a safe way to contribute, listen, then adopt, adapt or defer the idea with a reason. Revisit a deferred decision at an agreed time. Week 1 choices can change a program message or help arrangement. Week 2 mission and feature ideas feed Week 3 layout or Week 7 mechanics.

An adult response belongs in the learning conversation. Pupils do not need to obtain an adult signature to count their contribution. Staff keep the official AQA summary sheet separately from pupil voice and classroom response records.

GROW routines and evidence

Predictable stages with real pupil action and careful assessment

Stage and time	Routine
Arrival 0-3	Settle, open the right file and see one first action. Offer a quiet help signal and a clear return point after a break.
Starter 3-6	Retrieve one idea. Everyone predicts by speaking, pointing, showing a card or writing. Do not reveal the answer first.
I Do 6-11	Model one small action. Say what you notice and why you choose the block. Keep the relevant code visible.
We Do 11-18	Pupils predict, direct or perform the next step. Run the test, compare the result and explain one change together.
I Do 2 18-21	Model the next small skill or a repair. Connect it explicitly to the shared test.
We Do 2 · Task 21-25	Try the first task step together. Check understanding and choose a route before pupils work more independently.
Independent Task 25-35	Use a flexible route. Fade a prompt after success, change route when needed and observe each pupil's own contribution.
Exit 35-40	Save, reopen and show one result. Record pupil voice and give an adult response or a specific follow-up time.

Assessment and additional time

Baseline checks in Weeks 1-2 cover opening, editing or constructing, explaining, testing and saving. During the maze build, observe each required action and complete the official summary sheet. Record pupil ID, date, outcome, observed action, support, saved-file reference and next step. HTML predictions and screenshots can support a record but do not establish authorship alone.

At Week 2, agree extra foundation time if access or event/action understanding still blocks participation. At Week 6, review the first complete run and required scripts before adding complexity. Preserve time for the compulsory mechanic and tests. Extra lessons keep the existing unit outcomes; a change to award outcomes uses the school's AQA process in advance.

No pupil account or public sharing is needed for the planned local Scratch workflow. Save .sb3 projects in school-approved folders and check that pupils can reopen them.

Scoring continuation

Four further 40-minute sessions • AQA 122058 • Level One

[AQA 122058 Adding a scoring mechanism in Scratch](#)

Build on a working maze. Use two genuine scores: DeliveryPoints for completing delivery events and DiscoveryPoints for finding optional objects. Explain both before coding.

Week	Sequence and product	Assessment
9	Predict how each score changes. Model creating DeliveryPoints, then create DiscoveryPoints together. Pupils add and initialise both variables.	Explain what each variable measures and why both reset.
10	Model one event-linked update. Predict a second distinct event, construct its update and test both from reset.	Demonstrate at least two events that update score variables.
11	Investigate a pickup that credits repeatedly. Model hiding/deactivating after collection, then refine the pupil's own rule.	Identify a problem, implement a refinement and show its effect.
12	Agree a test set: new game, each scoring event, repeat contact and completed play. Pupils test and explain their revised system.	Review all four official outcomes using the summary sheet.

Full unit coverage

Plan for adding/managing scoring, knowledge of at least two score-tracking variables, knowledge of updates linked to at least two different game events, and experience implementing/refining scoring. Check the exact current outcome wording before sign-off.

40 minute structure for each continuation lesson

Arrival (0-3): reopen. Starter (3-6): predict. I Do (6-11) and We Do (11-18): model one scoring idea and test it together. I Do 2 (18-21) and We Do 2 · Task (21-25): model a second case and agree a test. Independent Task (25-35): construct or refine. Exit (35-40): save, explain and respond to pupil voice.

Support stretch and readiness

Support: start with one simple scoring event, use two distinct variable names, and keep a visible rule card. Stretch: explain repeated-credit prevention or compare fair scoring rules. If the underlying game is still unreliable, finish the core first.

All four outcomes use the summary sheet. Score plus lives is not assumed to satisfy the requirement for two score variables. Keep the original completed maze as a backup.

Optional route mission guide

AQA 119195 Using variables in Scratch • Level One

[AQA 119195 Using variables in Scratch](#)

Four sessions after basic Scratch, or five to six with extra foundation work. Product: a fictional guide that asks questions, remembers an earlier answer and uses it in a conversation.

Week	Teaching and application	Check
1	Predict an ask-and-answer script. Ask a fictional question, speak the answer, then join the answer with other text.	Pupil asks, speaks answer and joins it with a word or phrase.
2	Create a meaningful variable such as MissionName. Assign the answer to it, then ask a different question.	Pupil shows that stored MissionName survives the next answer.
3	Speak the stored variable and use it in a short conversation. Predict which value each speech block will show.	Pupil constructs a conversation with a stored variable.
4	Test different fictional inputs and repair a mix-up between answer and the stored variable. Explain names and the meaning of a variable.	Observe the practical outcomes and both knowledge outcomes.

Coverage checklist all nine outcomes

Practical outcomes	Knowledge outcomes
Ask a question; speak the answer; join the answer with other text; create a sensibly named variable; assign the answer to it; speak its stored value; create a conversation.	Explain why meaningful variable names help. Explain what a variable is.

The summary sheet is the published evidence requirement for all nine outcomes. Use fictional names and destinations so the activity does not need personal information.

Lesson structure and access

Use the core 40-minute rhythm. Retrieve the previous question at arrival, predict during the starter, model one block relationship, construct it together, then let pupils build a short version and save. Provide sentence choices rather than requiring extensive typing.

The key misconception is that answer and a named variable are always the same. After a second question, the answer reporter changes. The named variable keeps its assigned value until the program changes it. Use a two-question test to make this visible.

Stretch: add a second named variable and explain which question changes which stored value. Novice foundation sessions rehearse an event, the Looks category and saving before starting the four-session sequence.

Optional route three page website

AQA 112907 Making a basic webpage • Level One • Eight sessions

[AQA 112907 Making a basic webpage](#)

Product: a local interest guide, fictional club site or mini exhibition. Use an approved editor and preserve HTML and images in one project folder. The unit permits a suitable authoring tool; hand-written HTML/CSS adds a coding focus if appropriate.

Week	Focus / pupil product	Outcome alignment
1	Choose audience and topic, open the editor and create a titled homepage.	1, 2
2	Write short homepage text and add two readable font styles.	3, 4
3	Add at least two homepage images and check their display.	5
4	Create a titled first subpage with two pictures and short text.	6, 7
5	Add a Return to home button on that subpage. Test it.	8
6	Create a titled second subpage with one to five images.	9, 10
7	Add a hyperlink to a URL. Test navigation and assets from the saved folder.	11
8	Improve readability, repair broken links and save the complete site.	12 and review all

Evidence and exact requirements

Outcomes 1, 11 and 12 use the summary sheet. Outcomes 2-10 allow student completed work and/or summary sheet. Keep all three pages and required images. A single homepage does not meet the unit. The first subpage needs a Return to home button, not merely unlabelled navigation.

Teaching and access

Keep the same 40-minute rhythm. Model one page feature, build it together, predict what a link will open, then let pupils add content and test. Provide short text choices and a small set of original or appropriately licensed images. An HTML route can use a simple starter structure while pupils create the assessed page features.

Support: image folder already organised, visible file names and short content. Stretch: semantic headings, useful alternative text and consistent navigation. Avoid spending a session on accounts or public publishing; a saved local site is sufficient for this planned product.

Assessment must follow the published outcomes even when additional accessibility and design goals are taught.

Optional feedback and Python routes

Choose these after reviewing learner readiness

Sound and visual feedback 2 to 3 sessions

[AQA 122060 Adding sounds and visual effects in Scratch](#)

AQA 122060, Level One. Start with a working game. Plan two sounds and two visual effects so the design does not rely on an ambiguous combined minimum.

Week	Learning and evidence
1	Compare purposeful and distracting feedback. Model one sound and visual change, then pupils implement their selected effects.
2	Explain at least two methods of adding feedback and how choices affect the playing experience. Test with a chosen partner or teacher.
3 if needed	Refine the implementation after feedback, retest and explain the improvement.

All four outcomes use the summary sheet: enhancement, knowledge of methods, impact on experience and implementation/refinement. Use an agreed volume and visible equivalents for important messages. Avoid flashing effects. Preserve a working original.

Python bridge future 6 to 8 sessions

A possible next course for pupils who can independently save work, explain events/variables and tolerate text-based debugging. This is a curriculum bridge; no specific current desktop Level Two AQA unit is claimed for it.

Week	Possible progression
1-2	Run and edit a short print program. Read simple errors and save a file.
3-4	Use input and named variables in a fictional guide. Predict stored values.
5-6	Use one if/else choice and test both branches. Build a small text interaction.
7-8 if suitable	Add a bounded repeat or scoring rule, then test and explain an improvement.

Before commissioning the Python packs, verify a current unit against the exact planned outcomes and available software. Hardware-dependent Level Two units are not assumed suitable for a laptop-only course. If creating or amending a UAS unit is needed, agree this with the school lead before teaching towards the revised award.

Optional route microbit investigation

AQA 121956 • Level One • Six to eight sessions with devices

[AQA 121956: Coding and evaluating three types of step counter using BBC micro:bit](#)

Product: three working step-counter programs, comparable measurements and a reasoned evaluation. Physical micro:bit devices are necessary for the practical measurement elements. A simulator alone does not provide the complete experience.

Sessions	Teaching and practical task	Assessment focus
1-2	Code the standard step counter. Use a set distance and repeated measurements to calculate average steps.	Working code and the specified practical measurement.
3-4	Code the low-energy version. Repeat comparable measurements over the same set distance.	Second program, recorded values and average.
5-6	Code the sensitive version. Explore how acceleration strength affects its response.	Third program and investigation of sensitivity.
7-8 if needed	Evaluate all three versions. Explain shake and acceleration blocks and complete evidence.	Written evaluation and required knowledge evidence.

Resources and access

Check device numbers, connection method, batteries, approved coding environment and a measured safe route. Plan suitable participation in line with individual physical access arrangements. Keep the comparison fair by controlling the route and recording the test conditions.

Evidence requirements

The unit includes written work for the evaluation and two knowledge outcomes. Confirm each published evidence type for code and measurements in the live unit before preparing the final recording sheets. Do not replace the required practical use with a purely simulated activity.

Support and challenge

Support: one version at a time, a shared data table and a worked mean calculation. Each pupil still needs their own assessed contribution. Stretch: explain differences in sensitivity and identify why two devices or testers may produce different counts.

Decision before this route

Choose it only when devices, practical access and enough time are available. If the class needs a software-only next step, the scoring or variables route follows the main course more directly.

Adaptations for continuation routes

Use the same GROW stages and voice response routine in each later lesson

These adaptations change access and challenge while preserving each optional unit's full outcomes and evidence requirements. Select the route for the current skill; do not assign a pupil permanently to one column.

Route	Step by step	Main task	Challenge
Scoring	Use two named score cards and a physical counter. Build and test one event, then the second.	Create two real score variables, initialise them and test updates from two events.	Diagnose repeated credit and compare two fair point rules using test results.
Variables guide	Use fictional sentence choices and two labelled value spaces: answer and MissionName. Build one relation at a time.	Ask, store and reuse an answer in a conversation; test a second question.	Use another named variable; explain exactly which question changes each value.
Three-page site	Use an organised image folder, short text choices and a page checklist. Pupils create the required page features.	Create all three pages, required images, two font styles, Return to home button and URL link; save and test.	Improve semantic headings, useful image text and navigation; explain a usability test result.
Sound and effects	Compare one effect at a time. Choose from a small set with visible equivalents; build the required set over sessions.	Add two sounds and two visual effects, explain two methods and refine after a test.	Compare two feedback designs and justify the clearer or more comfortable choice.
Step counters	Use one version at a time, a prepared measurement table and worked mean example. Keep an individual contribution.	Code all three versions; make required practical comparisons and written evaluation.	Explain sensitivity differences and identify a variable that could make a comparison unfair.
Python bridge	Use one short line at a time, a word bank and an error pointer. Rehearse saving.	Build and test a small input, variable and selection program after foundation work.	Test both branches and a difficult input; explain one diagnostic repair.

Decisions before teaching a later route

Choose the product, software, access resources and number of sessions in advance. Keep the published evidence type: the micro:bit route needs physical use and specified written evidence. An adult may scribe exact pupil words where the centre permits; an oral answer alone must not silently replace a required written record.

The Python sequence is a possible curriculum bridge. Verify a suitable current AQA unit before preparing an award-specific pack; no desktop Level Two award is promised here.

Sources and implementation record

Links checked for this planning on 9 September 2026

Official AQA sources establish unit wording, level and evidence. Timings and classroom activities in this scheme are local planning estimates. Review the live unit again before an award submission.

[AQA 71638 Programming with Scratch unit 6](#)

[AQA 122058 Adding a scoring mechanism in Scratch](#)

[AQA 119195 Using variables in Scratch](#)

[AQA 112907 Making a basic webpage](#)

[AQA 122060 Adding sounds and visual effects in Scratch](#)

[AQA 121956 Comparing three step counters](#)

[AQA Unit Award Scheme](#)

[AQA UAS terms of business](#)

[AQA Our units](#)

Teaching and software references

[Scratch editor](#)

[Scratch offline app](#)

[Raspberry Pi Foundation PRIMM research](#)

[NCCE Supporting learners with SEND](#)

The prediction, investigation and construction sequence draws on PRIMM-informed computing teaching. Access arrangements use clear steps, repeated routines and flexible response formats; they require adaptation to the actual class.

School implementation decisions

Decision	Local record
Class / group and start date	
Scratch access and approved save location	
Week 2 readiness review / extra sessions	
Week 6 review / extra build and test time	
UAS lead and evidence arrangements	
Chosen continuation or alternative route	